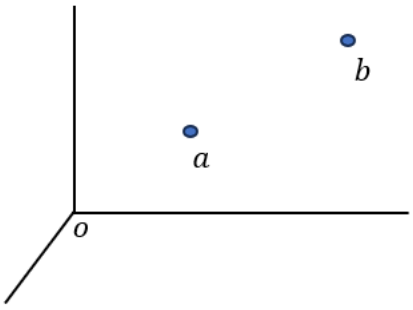
Potential energy

Consider any two points in a conservative field. The potential energy difference between the two points is defined as



$$U_b - U_a = -\int_{r_a}^{r_b} \vec{F}.d\vec{r}$$

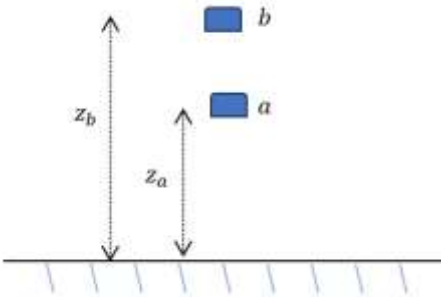
where U_b and U_a are the potential energies at points b and a, respectively. And r_a and r_b are the distances of a and b from origin of a chosen coordinate sysyem.

The above equation is valid only for a conservative force $\vec{F}$. Thus, *the potential energy difference between two points in a conservative field is a negative of the work done between the two points*.

We must also know that it is not possible to calculate the potential energy at a point. Potential energy at point has no meaning, as it has no unique value i.e., we can add any constant value to U_a , and same constant to U_b and still the difference remains the same. What we are interested is the *change in the potential energy (i.e., potential energy difference).*

Let us do some examples

1. Potential energy of a uniform force field
   Let us consider a particle of mass m in a uniform gravitational field $\vec{F} = -mg\hat{k}$. The change in potential energy when the particle move vertically (i.e., along z axis) from a to b is



$$\begin{aligned} U_b - U_a &= -\int_{z_a}^{z_b} \vec{F}.d\vec{z} \\ &= -\int_{r_a}^{r_b} (-mg\hat{k}).dz\hat{k} \ , \\ &\qquad \hat{k}\ is\ a\ unit\ vector\ alone\ z-axis. \\ &= mg\int_{r_a}^{r_b} dz \\ &= mg(z_b - z_a) \end{aligned}$$

   In most of the calculations we take the potential at the ground level to be zero i.e., $U_a = 0$ at $z = 0$. In this case we will write z_b as h, and U_b as $U(h)$. Therefore, from the above equation, we get

$$U(h) = mgh$$

   We often have a misconception in the above equation. In high school we often called $U(h) = mgh$ as the potential of the particle at height h, but actually it is the potential energy difference between the point and the ground level.

2. Potential energy of an inverse square force

If the force $f(r) \propto \frac{1}{r^2} = \frac{A}{r^2}$, where A is a constant, we called inverse square force. Examples of inverse square forces are gravitational force, electrostatic force, etc.

Let us consider a central force which is inverse square force. A force is called a central force if $\vec{F} = f(r)\hat{r}$. Therefore,

$$\begin{aligned} U_b - U_a &= -\int_{r_a}^{r_b} \vec{F}.d\vec{r} \\ &= -\int_{r_a}^{r_b} f(r)dr \\ &= -\int_{r_a}^{r_b} \frac{A}{r^2}dr \\ &= A\left(\frac{1}{r_b} - \frac{1}{r_a}\right) \end{aligned}$$

In most of the calculations we take the potential at the infinity to be zero i.e., $U_a = 0$ at $r = \infty$. So, now we simply write r_b as r and U_b as $U(r)$.

$$U(r) = \frac{A}{r}$$

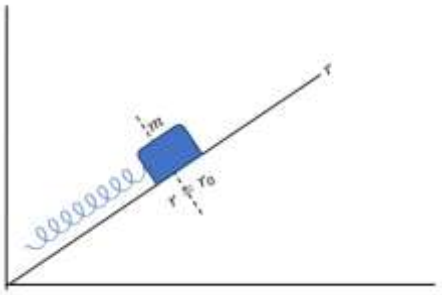
This gives the potential energy at r. But remember, it is with respect to infinity.

3. Linear harmonic oscillator

Let us consider a spring of equilibrium length r_0 with one end attached at the origin. If the spring is stretched to a length r along its length, then the force exerted on the spring is

$\vec{F} = -k(r - r_0)\hat{r}$,

where k is a spring constant.



Since the force is central, it is conservative.

The potential energy is

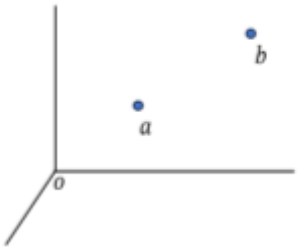
$$\begin{aligned} U(r) - u(r_0) &= -\int_{r_0}^{r} \vec{F}.d\vec{r} \\ &= -\int_{r_0}^{r} -k(r - r_0)dr \\ &= \frac{1}{2}k(r - r_0)^2 \end{aligned}$$

Like all other cases, we are free to chose our reference potential energy. We will take potential energy at the equilibrium to be zero i.e., $U(r_0) = 0$.

$$U(r) = \frac{1}{2}k(r - r_0)^2$$

Conservation of mechanical energy

Let us consider two points a and b in a conservative field.



We have, from Work-energy theorem,

$$K_b - K_a = W_{ba} = \int_{r_a}^{r_b} \vec{F}.d\vec{r} \qquad 1$$

And from potential energy,

$$U_b - U_a = -\int_{r_a}^{r_b} \vec{F}.d\vec{r} \qquad 2$$

Comparing the above two equations from eqn (1) and (2), we get

$$K_b - K_a = -(U_b - U_a)$$
$$or, \quad K_a + U_a = K_b + U_b \qquad 3$$

The left hand side of eqn (3) depends on the speed of the particle and potential energy at r_a, and make no reference to r_b. Similarly, the right hand side of eqn (3) depends on the speed of the particle and potential energy at r_b, and make no reference to r_a. This can be true if each sides of the equation equals to a constant, say E. Therefore,

$$K_a + U_a = K_b + U_b = E$$

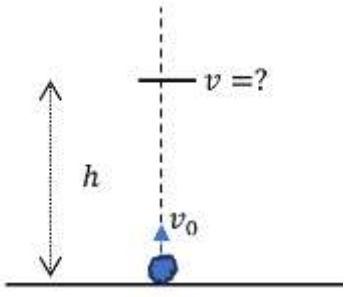
where E is called the total mechanical energy (simply called total energy) of the particle. In the language of physics, we called *energy is conserved*. This is the conservation of mechanical energy. This law has no new physical content, it simply presents a different ways of looking at a physical process. It is basically derived from Newton's law. It helps us to easily solve physical problems without directly applying Newton's law.

Examples

1. A mass projected upward with a given initial velocity in a constant gravity.
   Let an object of mass m be projected upward with an initial velocity v_0 in a constant gravity. Find the speed of the object at height h.

   Ans: By conservation of total mechanical energy,

   $$K_a + U_a = K_b + U_b$$
   $$or, \quad \frac{1}{2}mv_0^2 + U_0 = \frac{1}{2}mv^2 + mgh$$
   $$or, \quad \frac{1}{2}v_0^2 + 0 = \frac{1}{2}v^2 + gh$$
   $$or, \quad v = \sqrt{v_0^2 - 2gh}$$

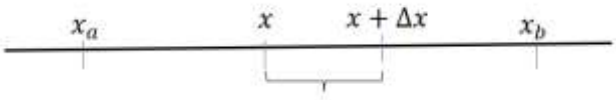


Force as gradient of potential energy

Let us consider a one dimensional motion, say along x axis. The potential energy of the particle between two points a and b is

$$U_a - U_b = -\int_{x_a}^{x_b} F(x)dx$$

Consider a small change in potential energy ΔU as the particle moves from point x to $x + \Delta x$.



$$U(x + \Delta x) - U(x) = -\int_{x}^{x+\Delta x} F(x)dx \qquad 1$$

For a small Δx, $F(x)$ can be considered constant over the range of integration. Therefore, from eqn (1), we have

$$\Delta U = -F(x)\int_{x}^{x+\Delta x} dx$$

$$\Delta U = -F(x)[x]_{x}^{x+\Delta x}$$

$$\Delta U = -F(x)\{x + \Delta x - x)$$

$$\Delta U = -F(x)\Delta x$$

In the limit $\Delta x \to 0$, we have

$$F(x) = -\frac{dU}{dx}$$

Thus, force is a negative gradient of potential energy. If the potential energy of the particle is known, the force acting on the particle can be found.

Stability

The result $F(x) = -dU/dx$ is useful not only for finding the force, but also for visualizing the stability of the system from the diagram of the potential energy. If the potential energy is given, we can analyse the stability of the system. This can be done in two ways-

a. Applying mathematical method.
b. From the graph of the potential energy.

Mathematical method

Consider a point x_0. If $\frac{dU}{dx}|_{x_0} = 0$, then the point x_0 is called an equilibrium point. That is, the net force on the particle is zero at equilibrium. Equilibrium point can be stable or unstable. To find out whether it is stable or unstable, we have to do the followings.

- If $\frac{d^2U}{dx^2}|_{x_0} > 0$, then x_0 is stable equilibrium point.
- If $\frac{d^2U}{dx^2}|_{x_0} < 0$, then x_0 is unstable equilibrium point.
- If $\frac{d^2U}{dx^2}|_{x_0} = 0$, then we need to look for higher derivatives. If all higher derivatives are zero at x_0, the point x_0 is a neutral stability point.

Example 1.

A particle moves along the x axis in a force field having potential $U = \frac{1}{2}kx^2$ with $k > 0$.

(a) Determine the points of equilibrium.

(b) Find the stability.

Ans:

$$U = \frac{1}{2}kx^2, \qquad k > 0$$

Now,

$$\frac{dU}{dx} = kx$$

(a) $\frac{dU}{dx}$ becomee zero at $x = 0$. Therefore, $x = 0$ is the equilibrium point.

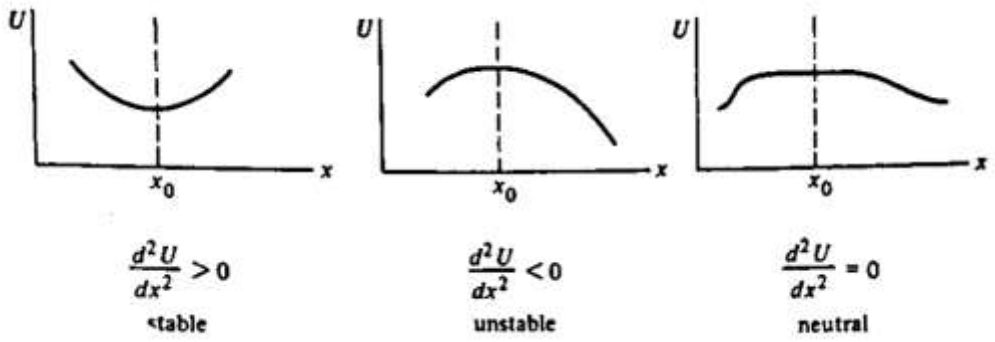
(b) To find the stability, we have to take second derivative.

$$\frac{d^{2U}}{dx^2}|_{x_0} = k > 0$$

Hence, $x = 0$ is stable equilibrium.

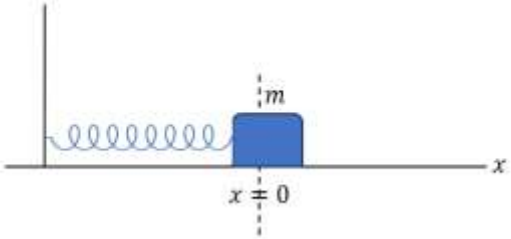
From the graph of the potential energy.

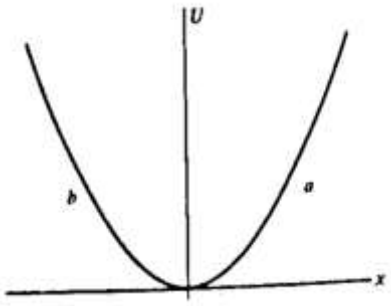
Draw the graph of U as a function of position. The points where the graph is maximum gives unstable point, and the minimum gives stable point.



Examples

1. Harmonics oscillator





Potential energy is $U(x) = \frac{1}{2}kx^2, k > 0$

To check the stability, we have to draw the graph.

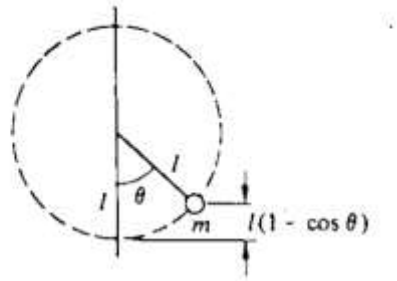
From the diagram, stable point is $x = 0$.

2. Pendulum

Potential energy is

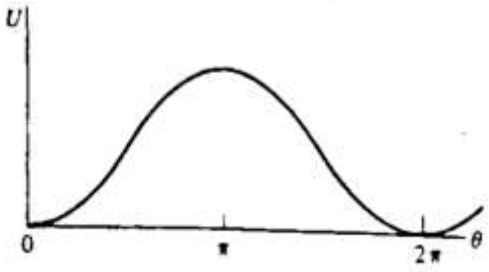
$$U(\theta) = mgz = mgl(1 - cos\theta)$$

Taking potential energy to be zero at the bottom of the swing.



To check the stability, we have to draw the graph.

From the diagram, the bottom i.e., $\theta = 0$ is stable. But the top i.e., $\theta = \pi$ is unstable.

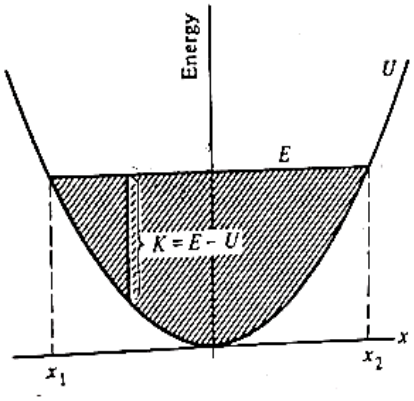


Energy diagrams

We can often find the most interesting features of the motion of a system (system may be a particle, etc.) by using *energy diagram*. A diagram in which total energy E and potential energy U is plotted as a function of position is called energy diagram.

Kinetic energy is $K = E - U$. [Because $E = K + U$]

Since kinetic energy K can not be negative, the motion of the system is constraint to region where $U \leq E$. [$K \geq 0$ is possible only when $U \leq E$]



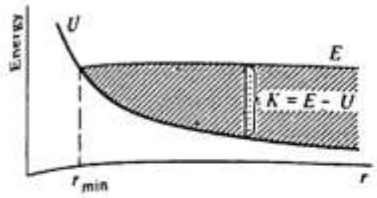
4.9 Energy Diagrams

We can often find the most interesting features of the motion of a one dimensional system by using an *energy diagram*, in which the total energy E and the potential energy U are plotted as functions of position. The kinetic energy $K = E - U$ is easily found by inspection. Since kinetic energy can never be negative, the motion of the system is constrained to regions where $U \leq E$. Here is the energy diagram for a harmonic oscillator. The potential energy $U = kx^2/2$ is a parabola centered at the origin. Since the total energy is constant for a conservative system, E is represented by a horizontal straight line. Motion is limited to the shaded region where $E \geq U$; the limits of the motion, x_1 and x_2 in the sketch, are sometimes called the turning points.

Here is what the diagram tells us. The kinetic energy, $K = E - U$, is greatest at the origin. As the particle flies past the origin in either direction, it is slowed by the spring and comes to a complete rest at one of the turning points x_1, x_2. The particle then moves toward the origin with increasing kinetic energy, and the cycle is repeated.

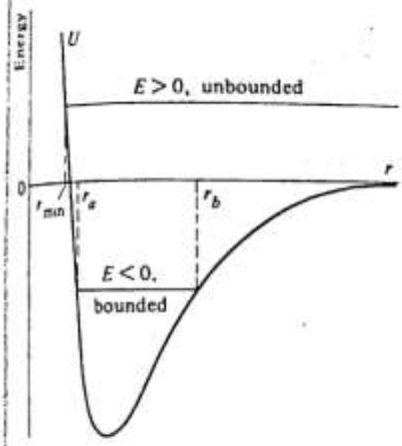
The harmonic oscillator provides a good example of bounded motion. As E increases, the turning points move farther and farther off, but the particle can never move away freely. If E is decreased, the amplitude of motion decreases, until finally for $E = 0$ the particle lies at rest at $x = 0$.

Quite a different behavior occurs if U does not increase indefinitely with distance. For instance, consider the case of a particle constrained to a radial line and acted on by a repulsive inverse



square law force $A\bar{r}/r^2$. Here $U = A/r$, where A is positive. There is a distance of closest approach, r_{min}, as shown in the diagram, but the motion is not bounded for large r since U decreases with distance. If the particle is shot toward the origin, it gradually loses kinetic energy until it comes momentarily to rest at r_{min}. The motion then reverses and the particle moves out toward infinity. The final and initial speeds at any point are identical; the collision merely reverses the velocity.

With some potentials, either bounded or unbounded motion can occur depending upon the energy. For instance, consider the interaction between two atoms. At large separations, the atoms attract each other weakly with the van der Waals force, which varies as $1/r^7$. As the atoms approach, the electron clouds begin to overlap, producing strong forces. In this intermediate region the force is either attractive or repulsive depending on the details of the electron configuration. If the force is attractive, the potential energy decreases with decreasing r. At very short distances the atoms always repel each other strongly, so that U increases rapidly as r becomes small.



The energy diagram for a typical attractive two atom system is shown in the sketch. For positive energy, $E > 0$, the motion is unbounded, and the atoms are free to fly apart. As the diagram indicates, the distance of closest approach, r_{min}, does not change appreciably as E is increased. The steep slope of the potential energy curve at small r means that the atoms behave like hard spheres—r_{min} is not sensitive to the energy of collision.

The situation is quite different if E is negative. Then the motion is bounded for both small and large separations; the atoms never approach closer than r_a or move farther apart than r_b. A bound system of two atoms is, of course, a molecule, and our sketch represents a typical diatomic molecule energy diagram. If two atoms collide with positive energy, they cannot form a molecule unless some means is available for losing enough energy to make E negative. In general, a third body is necessary to carry off the excess energy. Sometimes the third body is a surface, which is the reason surface catalysts are used to speed certain reactions. For instance, atomic hydrogen is quite stable in the gas phase even though the hydrogen molecule is tightly bound. However, if a piece of platinum is inserted in the hydrogen, the atoms immediately join to form molecules. What happens is that hydrogen atoms tightly adhere to the surface of the platinum, and if a collision occurs between two atoms on the surface, the excess energy is released to the surface, and the molecule, which is not strongly attracted to the surface, leaves. The energy delivered to the surface is so large that the platinum glows brightly. A third atom can also carry off the excess energy, but for this to happen the two atoms must collide when a third atom is nearby. This is a rare event at low pressures, but it becomes increasingly important at higher pressures. Another possibility is for the two atoms to lose energy by the emission of light. However, this occurs so rarely that it is usually not important.